NPTEL Software Testing

100 NPTEL-Style Practice MCQs - Graph Questions Included

Single-correct-answer practice set. Questions 21-50 emphasize graph testing, including explicit graph/CFG problems. The answer key is only on the final page.

1. A defect in the source code is called a:

A. failure
B. fault
C. oracle
D. test path

2. An incorrect internal program state produced during execution is an:

A. error
B. edge
C. input block
D. test requirement

3. In RIPR, which condition means the faulty statement must execute?

A. Reachability
B. Infection
C. Propagation
D. Revelation

4. A faulty expression is executed, but it produces the same value as the correct expression. Which RIPR condition fails?

A. Reachability
B. Infection
C. Propagation
D. Revelation

5. A wrong internal state is created but overwritten before output. Which RIPR condition fails?

A. Reachability
B. Infection
C. Propagation
D. Revelation

6. A wrong output is produced but the test has no assertion checking it. Which RIPR condition fails?

A. Reachability
B. Infection
C. Propagation
D. Revelation

7. A test oracle primarily determines:

A. the control-flow graph
B. whether observed behavior passes or fails
C. the number of mutants
D. the program language

8. Which statement about structural coverage is most accurate?

A. 100% coverage proves correctness
B. Coverage gives evidence about executed obligations but faults may remain
C. Coverage removes the need for oracles
D. Coverage guarantees all mutants are killed

9. For an input partition to be complete:

A. every block must have equal size
B. every possible input must belong to some block
C. blocks must overlap at boundaries
D. all blocks must be valid inputs

10. For an input partition to be disjoint:

A. no input belongs to more than one block
B. every block has one input
C. all blocks are numeric
D. every input is valid

11. Interface-based input modelling mainly starts from:

A. individual parameters at the interface
B. only program outputs
C. only control-flow paths
D. mutation operators

12. Functionality-based input modelling mainly starts from:

A. requirements and system behavior
B. variable names only
C. compiler errors
D. CPU utilization

13. Three characteristics have 2, 3 and 4 blocks. How many combinations does ACoC require?

A. 9
B. 12
C. 18
D. 24

14. Each Choice Coverage requires:

A. every cross-product combination
B. every block of every characteristic at least once
C. every pair of graph edges
D. one base test only

15. Pair-wise coverage requires:

A. every test to contain two inputs
B. every 2-way combination of blocks across characteristic pairs
C. exactly two tests
D. all complete paths

16. If t equals the number of characteristics, t-wise coverage becomes:

A. ECC
B. BCC
C. ACoC
D. Node Coverage

17. Base Choice Coverage is built by:

A. changing all blocks simultaneously
B. choosing a base test and changing one characteristic at a time
C. using only invalid blocks
D. executing every path

18. Characteristics have 3, 2 and 4 blocks. Ignoring infeasibility, BCC requires how many tests?

A. 7
B. 8
C. 9
D. 24

19. A combination of blocks that cannot occur because of constraints is called:

A. prime
B. infeasible
C. equivalent
D. def-clear

20. Which technique is most directly aimed at reducing a large cross-product while retaining 2-way interactions?

A. Pair-wise coverage
B. All Combinations Coverage
C. Complete Path Coverage
D. All-DU-Paths

21. In graph terminology, a test path normally:

```
Graph G1:

    1
    |
    2
   / \
  3   4
   \ /
    5
```

A. starts at an initial node and ends at a final node
B. must be a cycle
C. contains every edge
D. cannot contain repeated nodes

22. For G1, which set of test paths achieves Node Coverage?

```
Graph G1:

    1
    |
    2
   / \
  3   4
   \ /
    5
```

A. {1,2,3,5} only
B. {1,2,4,5} only
C. {1,2,3,5} and {1,2,4,5}
D. {2,3,5} only

23. For G1, which requirement is NOT satisfied by only the path [1,2,3,5]?

```
Graph G1:

    1
    |
    2
   / \
  3   4
   \ /
    5
```

A. Visiting node 1
B. Visiting node 2
C. Visiting node 4
D. Traversing edge (3,5)

24. For G1, Edge Coverage requires which additional edge after test path [1,2,3,5] has run?

```
Graph G1:

    1
    |
    2
   / \
  3   4
   \ /
    5
```

A. (1,2)
B. (2,3)
C. (3,5)
D. (2,4) and (4,5)

25. An edge-pair is:

A. one node
B. one edge
C. a path containing two consecutive edges
D. every edge in the graph

26. Which sequence is an edge-pair in G2?

```
Graph G2 (edges):
1 -> 2
2 -> 3
3 -> 2
2 -> 4
4 -> 5
```

A. [1,2]
B. [1,2,3]
C. [1,2,4,5]
D. [2]

27. For G2, [2,3,2] is best described as:

```
Graph G2 (edges):
1 -> 2
2 -> 3
3 -> 2
2 -> 4
4 -> 5
```

A. a simple cycle/round trip
B. a complete test path from 1 to 5
C. an infeasible sequence
D. a single edge

28. Why can Complete Path Coverage be impossible to satisfy finitely for G2?

```
Graph G2 (edges):
1 -> 2
2 -> 3
3 -> 2
2 -> 4
4 -> 5
```

A. there is no final node
B. the 2-3-2 loop can be repeated arbitrarily many times
C. node 1 has no edge
D. all paths are simple

29. A simple path:

A. never repeats a node, except first and last may match for a cycle
B. must contain all nodes
C. cannot contain edges
D. must start at the initial node

30. A prime path is:

A. any path of length one
B. a maximal simple path
C. a path containing all edges
D. a path that always starts at node 1

31. Prime Path Coverage requires:

A. every reachable node only
B. every reachable edge only
C. every feasible prime path to be toured
D. all truth-table rows

32. In the common graph-coverage strength progression, which sequence is correct?

A. PPC -> EPC -> EC -> NC
B. NC -> EC -> EPC -> PPC
C. EC -> NC -> PPC -> EPC
D. NC -> PPC -> EC -> EPC

33. For G3, the path [1,2,3,5] and path [1,2,4,5] together cover:

```
Graph G3:
1 -> 2 -> 3 -> 5
     \-> 4 ->/
```

A. only node 2
B. all shown nodes and all shown edges
C. no edges
D. only one branch

34. A direct tour of required path [2,3,5] means the test path contains:

```
Graph G3:
1 -> 2 -> 3 -> 5
     \-> 4 ->/
```

A. [2,3,5] contiguously
B. nodes 2,3,5 in any order
C. only nodes 2 and 5
D. a detour replacing every edge

35. A sidetrip in graph testing generally:

A. leaves and returns while preserving the required route order
B. deletes a required edge
C. makes every path prime
D. forces a new initial node

36. A detour differs from direct touring because it can:

A. replace a required edge with another path between its endpoints
B. remove the final node
C. change all nodes into clauses
D. guarantee mutation killing

37. In G4, which edge is a back edge forming the loop?

```
Graph G4:
1 -> 2 -> 3
     ^    |
     |____|
2 -> 4 -> 5
```

A. (1,2)
B. (2,3)
C. (3,2)
D. (4,5)

38. In G4, which sequence represents one traversal of the simple cycle?

```
Graph G4:
1 -> 2 -> 3
     ^    |
     |____|
2 -> 4 -> 5
```

A. [1,2,4,5]
B. [2,3,2]
C. [3,2,4]
D. [1,2,3,2,3]

39. Which statement about Node Coverage and Edge Coverage is generally correct for a reachable CFG?

A. NC always subsumes EC
B. EC normally subsumes NC
C. they are always identical
D. neither involves test requirements

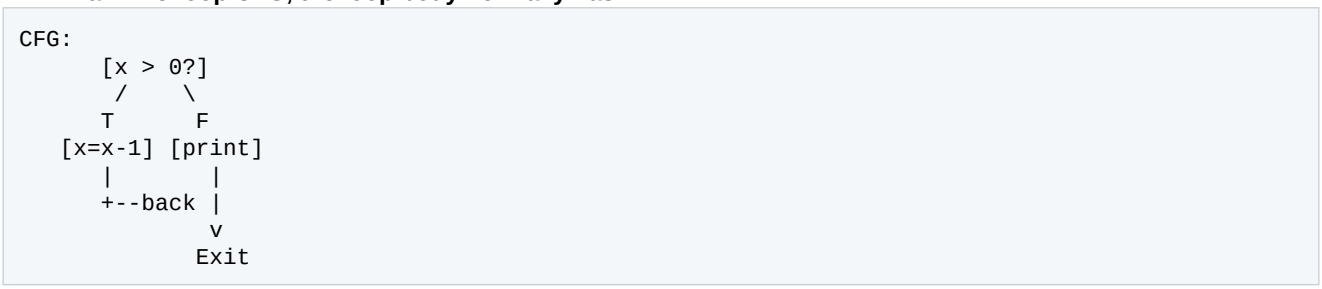
40. Edge-Pair Coverage can expose faults missed by Edge Coverage because it requires:

A. particular consecutive edge sequences
B. only initial nodes
C. all input combinations
D. all mutants

41. In a CFG, an if statement typically introduces:

A. a branch in control flow
B. a mutation score
C. a def-clear path automatically
D. a base choice

42. In a while-loop CFG, the loop body normally has:



A. no connection to the condition
B. a back edge toward the loop test/update
C. only one final node and no exit
D. no branch

43. A break inside a loop normally transfers control to:

A. the loop body start
B. the loop exit
C. the previous method
D. the condition with true forced

44. A continue inside a loop normally transfers control toward:

A. the next iteration condition/update
B. the method exit
C. the exception handler only
D. the previous statement only

45. A return statement in a CFG normally transfers control toward:

A. method exit
B. loop header only
C. another branch condition
D. the initial node

46. For an if-else CFG, achieving Edge Coverage generally requires:

A. taking both true and false branch edges when feasible
B. executing only the true branch
C. executing only the false branch
D. avoiding the decision node

47. If a required graph path is infeasible, a tester should:

A. invent an input anyway
B. recognize the obligation cannot be realized under the constraints
C. mark every test failed
D. convert it to a mutant

48. Which graph model represents possible method-to-method calls?

A. call graph
B. truth table
C. input partition
D. mutation table

49. Which graph model is most natural for protocol states and events?

A. finite-state machine
B. arithmetic expression tree only
C. input cross-product only
D. mutation score graph

50. Transition coverage in a finite-state machine is most analogous to:

A. edge coverage
B. clause coverage
C. all-defs
D. ECC

51. In data-flow testing, x=10 is a:

A. use
B. definition
C. predicate use only
D. final node

52. In y=x+1, the reference to x is a:

A. c-use
B. p-use
C. definition
D. mutant

53. In if(x>0), the reference to x is a:

A. c-use only
B. p-use
C. definition
D. base choice

54. A def-clear path from definition d to use u contains:

A. no redefinition of that variable between d and u
B. no branches
C. no uses
D. no edges

55. A du-pair connects:

A. two definitions
B. a definition with a reachable use of the same variable
C. two graph nodes regardless of variables
D. two clauses

56. Consider: 1:x=0; 2:if(a); 3:x=1; 4:print(x). Which path is def-clear for the definition at 1 reaching use at 4?

A. 1-2-3-4
B. 1-2-4
C. 3-4 only represents definition 1
D. no path can be def-clear

57. In the same code, why is 1-2-3-4 not def-clear for definition 1?

A. node 2 is a predicate
B. x is redefined at node 3
C. node 4 is a use
D. the path is too short

58. All-Defs Coverage requires:

A. at least one reachable use for every definition
B. every possible use for every definition
C. every simple def-clear path
D. every input value

59. All-Uses Coverage requires:

A. one use in the entire program
B. every reachable use for each definition
C. all prime paths only
D. all input pairs

60. All-DU-Paths Coverage requires:

A. every simple def-clear path for every du-pair
B. one use per definition only
C. all browser combinations
D. all mutants to compile

61. Which is the usual strength order from weaker to stronger?

A. All-DU-Paths -> All-Uses -> All-Defs
B. All-Defs -> All-Uses -> All-DU-Paths
C. All-Uses -> All-Defs -> All-DU-Paths
D. All-Defs -> All-DU-Paths -> All-Uses

62. A use-before-definition pattern is primarily a:

A. data-flow anomaly
B. pair-wise combination
C. load test
D. prime cycle

63. In P = A AND B, P is the:

A. clause
B. predicate
C. definition
D. mutant

64. In P = A AND B, A and B are:

A. clauses
B. test paths
C. definitions
D. mutants

65. Predicate Coverage requires P to evaluate:

A. true only
B. false only
C. both true and false
D. all 2^n clause combinations

66. Clause Coverage requires:

A. each clause true and false at least once
B. only the predicate true and false
C. all graph nodes
D. all inputs

67. For three independent clauses, Combinatorial Coverage has how many truth assignments?

A. 3
B. 6
C. 8
D. 9

68. For P=A OR B, when does A determine P?

A. when B=true
B. when B=false
C. for any B without restriction
D. never

69. The clause selected to demonstrate independent influence is the:

A. minor clause
B. major clause
C. base clause
D. final clause

70. GACC allows the minor-clause values in the paired tests to:

A. differ
B. never differ
C. all be true
D. all be false

71. CACC additionally requires:

A. the predicate outcomes to differ
B. the predicate outcomes to be identical
C. all clauses to be true
D. no feasibility constraint

72. RACC restricts the paired tests so that:

A. minor clauses are identical while the major clause changes
B. all clauses change
C. the predicate stays constant
D. no clause determines the predicate

73. For P=A OR B with A major, which pair satisfies RACC?

A. (A=T,B=F) and (A=F,B=F)
B. (T,T) and (F,F)
C. (T,T) and (F,T)
D. (T,F) and (T,T)

74. Inactive Clause Coverage focuses on cases where:

A. changing the major clause does not affect the predicate
B. every clause controls the predicate
C. all mutants are killed
D. all graph edges execute

75. MC/DC is most closely related to:

A. active clause coverage and independent condition influence
B. load testing
C. input partition completeness
D. all-defs

76. Which statement is correct?

A. CACC requires correlated predicate change
B. RACC lets all minors freely change
C. GACC always fixes minors
D. CoC requires only n tests

77. A mutant is killed when:

A. it compiles successfully
B. a test produces observably different behavior from the original
C. it is equivalent
D. it has no changed token

78. Changing > to >= is an example of:

A. relational operator mutation
B. node coverage
C. base choice
D. load testing

79. Changing + to - is an example of:

A. arithmetic operator mutation
B. clause coverage
C. state coverage
D. recovery testing

80. A surviving mutant may mean:

A. the tests are weak or the mutant is equivalent
B. the software is certainly correct
C. the test suite has 100% CoC
D. the CFG has no nodes

81. An equivalent mutant:

A. behaves identically to the original for all inputs
B. always crashes
C. can always be killed by boundary testing
D. is a graph edge

82. Mutation score should normally exclude confirmed:

A. killed mutants
B. equivalent mutants
C. test cases
D. operators

83. If 90 non-equivalent mutants exist and 81 are killed, the mutation score is:

A. 81%
B. 90%
C. 9%
D. 111%

84. Weak mutation typically checks infection:

A. near the mutated location
B. only after final output
C. only before reachability
D. only in requirements

85. Strong mutation requires the infected state to:

A. propagate to an observable difference
B. remain hidden
C. avoid execution
D. become a graph node

86. Statement deletion is an example of:

A. a mutation operator
B. a finite-state event
C. an input block
D. a regression prioritization method

87. Dynamic binding selects an overridden method primarily according to the:

A. runtime receiver type
B. source filename
C. test-case number
D. declared variable name only

88. A yo-yo effect is associated with:

A. deep inheritance and overriding
B. pair-wise ISP
C. soak testing
D. simple paths only

89. Polymorphism testing should consider:

A. relevant runtime receiver types
B. only declared types
C. only static pages
D. only equivalent mutants

90. Retest-all means:

A. rerun the entire existing test suite
B. run only failed tests
C. remove redundant tests
D. run only new tests

91. Regression test selection primarily:

A. chooses tests affected by changes
B. deletes every slow test
C. changes requirements
D. creates mutants

92. Regression test minimization primarily:

A. removes tests while preserving selected obligations
B. reorders without removing tests
C. adds all possible tests
D. changes runtime types

93. Regression test prioritization primarily:

A. reorders tests to obtain earlier feedback
B. guarantees no regression
C. removes every duplicate
D. performs mutation automatically

94. Load testing evaluates behavior under:

A. expected workload
B. only workload far beyond capacity
C. only sudden change
D. only long idle periods

95. Stress testing evaluates behavior:

A. beyond expected capacity
B. only at expected capacity
C. only when offline
D. only with one user

96. Which mapping is correct?

A. Spike=sudden load change; Soak=sustained load over time
B. Spike=sustained; Soak=sudden
C. Load=beyond capacity; Stress=expected
D. Soak=pair-wise input testing

97. Authentication answers:

A. Who are you?
B. What are you allowed to do?
C. How fast is the system?
D. Which path is prime?

98. Authorization answers:

A. Who are you?
B. What are you allowed to do?
C. How many blocks exist?
D. Which mutant survives?

99. For web applications, client-side validation should generally be backed by:

A. server-side validation
B. no validation
C. CSS only
D. browser cache only

100. Which mobile test is most important for lifecycle robustness?

A. background/foreground, rotation and process death
B. only changing font color
C. only browser cookies
D. only truth tables

Answer Key - Final Page

Check this page only after completing the full set.

1 B	**2** A	**3** A	**4** B	**5** C	**6** D	**7** B	**8** B	**9** B	**10** A
11 A	**12** A	**13** D	**14** B	**15** B	**16** C	**17** B	**18** A	**19** B	**20** A
21 A	**22** C	**23** C	**24** D	**25** C	**26** B	**27** A	**28** B	**29** A	**30** B
31 C	**32** B	**33** B	**34** A	**35** A	**36** A	**37** C	**38** B	**39** B	**40** A
41 A	**42** B	**43** B	**44** A	**45** A	**46** A	**47** B	**48** A	**49** A	**50** A
51 B	**52** A	**53** B	**54** A	**55** B	**56** B	**57** B	**58** A	**59** B	**60** A
61 B	**62** A	**63** B	**64** A	**65** C	**66** A	**67** C	**68** B	**69** B	**70** A
71 A	**72** A	**73** A	**74** A	**75** A	**76** A	**77** B	**78** A	**79** A	**80** A
81 A	**82** B	**83** B	**84** A	**85** A	**86** A	**87** A	**88** A	**89** A	**90** A
91 A	**92** A	**93** A	**94** A	**95** A	**96** A	**97** A	**98** B	**99** A	**100** A